

Hyperspectral target detection using cluster-based probability models implemented in a generalized likelihood ratio test

Itan Levin, Tomer Hershkovitz, Stanley Rotman

Ben-Gurion Univ. of the Negev (Israel), Department of Electrical and Computer Engineering

ABSTRACT

We present an algorithm for sub-pixel target detection in hyperspectral images, considering both the common additive target model, and a replacement target model where the target's spectrum partially replaces that of the background. We implement an LRT based decision rule, estimating the underlying distributions using cluster detection in feature subsets of a decorrelated image. We select these subsets in subspaces corresponding to sets of consecutive eigenvalues of the data's empiric covariance. The densities are approximated using products of lower dimensional Gaussian mixture models. We utilize the estimated density functions to compute maximum likelihood estimates of the target's relative fraction of the observed pixel spectrum, and obtain a GLRT based test statistic. Performance analysis of this proposed detector shows promising results when compared to the detection capabilities of the matched filter, which is used as a benchmark.

Keywords: Hyperspectral, Subpixel target detection, Replacement model, GLRT, PCA, GMM, unsupervised clustering.

1. Introduction

Hyperspectral optical sensors are capable of generating high dimensional datasets, with accurate characterization of the observed elements' reflectance or radiance spectra. In cases where the identification of a specific material or object is required and knowledge of its unique spectral signature is available, images obtained from such sensors offer a powerful detection tool.

In a detection problem framework, the Neyman-Pearson lemma states that for a given maximum false alarm rate, the likelihood ratio test (LRT) is most powerful, i.e. yields highest probability of detection. As a test statistic obtained from observation $\mathbf{x}$, The LRT is given by:

$$\Lambda(\mathbf{x}) = \frac{f(\mathbf{x}|H_1)}{f(\mathbf{x}|H_0)} \underset{H_0}{\overset{H_1}{\gtrless}} \eta \quad (1)$$

where η is the decision threshold, determined by the desired false alarm rate. Assuming an additive target and normally distributed clutter, this decision rule becomes equivalent to the commonly used and extensively studied matched filter (MF) detector. A sometimes appealing property of the MF is maximizing the signal to noise ratio (SNR) when an additive target is assumed.

Other than the normal distribution, various parametric distribution models may be used, to be substituted in (1) and obtain an explicit test statistic. Although appealing from a mathematical point of view, a convincing justification for using a specific parametric distribution model does not always exist. The generalized likelihood ratio test (GLRT) is used when a parametric model is utilized, but with missing parameter values, which are replaced by their maximum likelihood estimates.

This research is focused on the detection of subpixel targets with a known spectral signature within a hyperspectral data cube. We have based our detection algorithm on the GLRT, which requires the formulation of two competing hypotheses, and the estimation of an underlying probability distribution for each pixel being examined, under each of the two hypotheses. The null hypothesis in this case is that the requested target is not present in the observed

pixel, while the alternative hypothesis is that the target is present. Having estimated the distributions, the implemented decision rule takes on the form given in (1), where $\mathbf{x}$ is an observed pixel. Instead of complete parametric models, we use semi-parametric background distribution modelling, which takes into account the multimodal behavior commonly encountered in hyperspectral data. We will consider a replacement target model instead of an additive model, i.e., a scenario in which the background signature is partially replaced with the target signature. This is not uncommon in the case of opaque targets. We will also consider the need for estimating the fraction of the target signature in the pixel for both the additive and replacement model. Finally, we will consider wavelength dependent values of fractional target fill, due to wavelength dependent blur.

Estimation of the probability density function (PDF) under the null hypothesis is performed using the entire data cube, assuming the vast majority of pixels do not contain the desired target. In [2], an estimation method was proposed where the data is first decorrelated using principle component analysis, and an optimal spanning tree is generated to capture pairwise dependencies between different data features. We follow a somewhat similar approach, but turn to segment the decorrelated features according to their variance, and identify clusters in the data, fitting a Gaussian mixture probability distribution model as our PDF estimation.

2. Algorithmic Goals

2.1 Target and background modelling

Each pixel we analyze is seen as a vector in Euclidean N-dimensional space, where each element corresponds to the observed reflectance at a different wavelength. We assume it is drawn from some probability distribution which describes the background present in the image, and possibly the requested target. In order to implement an LRT based decision rule, mathematical models of the background and the target need to be determined, as well as the underlying distributions.

An intuitive mathematical model for the case where a target occupies some portion of a pixel, is the *replacement target model*:

$$\mathbf{x} = p\mathbf{t} + (1 - p)\mathbf{b}, 0 < p < 1 \quad (2)$$

Here $\mathbf{x}$ is an observed pixel, $\mathbf{t}$ is the requested target's spectrum, $\mathbf{b}$ is the background spectrum, and the parameter p describes the ratio between the target and background spectra present in the given pixel, sometimes referred to as the *fill factor*. Manolakis et al. [3] point out a limiting case which is of special interest:

$$\mathbf{x} \approx p\mathbf{t} + \mathbf{b}, \text{ when } 0 < p \ll 1 \quad (3)$$

This approximation describes the commonly used *additive target model*. As stated, it is a good approximation in cases where the background's spectrum is expected to be dominant.

In both cases, the null hypothesis (H_0) is $p = 0$, and the alternative hypothesis (H_1) is $p > 0$.

2.2 Wavelength dependent target fill factor

Other than the visible proportions of objects in the image, the factor p could be affected by the physical characteristics of the sensor and environmental conditions. The effect on the observed reflectance is not necessarily uniform over all different wavelengths, and a deformation of the spectrum may take place. To capture this, we extend the above models, so that p is replaced with a diagonal matrix P , whose elements correspond to the wavelength-specific fill factors. Equations (2) and (3) thus become:

$$\mathbf{x} = P\mathbf{t} + (I - P)\mathbf{b} \quad (4)$$

$$\mathbf{x} = P\mathbf{t} + \mathbf{b} \quad (5)$$

For sub-pixel targets, $0 < P < I$ is satisfied, and for the null hypothesis, P is the zero matrix.

2.3 Gaussian Mixture Model (GMM) Distribution

A basic probabilistic model for the image's background is the multivariate normal (MVN) distribution. This model has been extensively studied due to its relative simplicity. When a normal distribution is assumed, the LRT and GLRT decision rules result in simple, easily calculated expressions. The obtained detection algorithms include the matched filter (MF) [1], adaptive matched filter (AMF) [4] and adaptive coherence estimator (ACE) [5]. Several other examples exist, as well as variations to the ones given above [6] - [9].

The nature of images obtained by means of remote sensing is to contain the spectra of several different materials and land covers that are present in the observed region. Vegetation, different types of soil, bodies of water and urban regions can all appear in a single hyperspectral image and display very different, and clearly non stationary spectral characteristics. A Gaussian distribution, or generally any unimodal distribution will be very unlikely to capture these differences or offer a good statistical explanation of the observed data. The concept of spectral endmembers suggests that each distinct type of land cover may be described as a spectral class with its own statistical model. If we assume a linear mixing model, any pixel in the given image may be described as a linear combination of the endmember classes [10]:

$$\mathbf{x} = \sum_{i=1}^M a_i \mathbf{s}_i \quad (6)$$

where $\mathbf{x}$ is the received pixel spectrum vector, and each $\mathbf{s}_i$ in the above linear combination is drawn from a different endmember class distribution. a_i are the linear mixing coefficients, and $\mathbf{a} = (a_1, \dots, a_M)^T$ is also called the *abundance* vector.

Distributions of endmember classes in themselves are by no means guaranteed to be unimodal, as demonstrated in Figure 1.

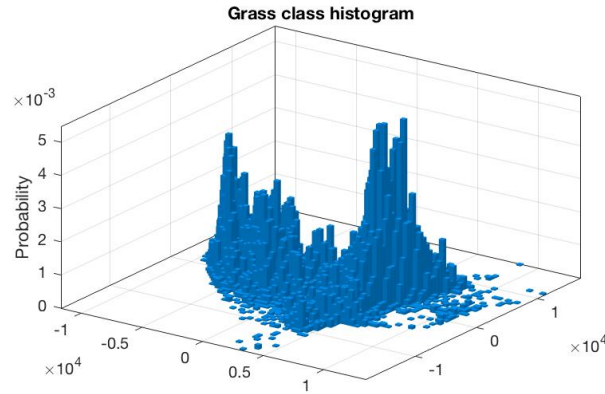


Figure 1: 2D Histogram generated using pixels of meadow regions containing only grass, taken from the GIC *indian pines* dataset. Reduction to two dimensions was done using truncated PCA. The first two dimensions, which were used to construct the histogram, retain 92% of the overall spectral information. Data distribution in this case is clearly multi-modal.

It is in fact generally difficult to obtain a complete parametric model which captures these spectral classes' underlying distributions accurately.

An alternative family of statistical models which has been pointed out [10] as having the potential to capture this multi-modal structure, is the family of mixture-model distributions. Specifically, The Gaussian mixture model (GMM) has been the subject of several studies in this context [11] – [14]. Its appeal is mainly due to a relatively simple iterative process which allows fitting such a model to given data, which can be derived via the Expectation-Maximization (EM) algorithm [15].

Zhou et al. have shown that under the assumption that each of the endmember classes are distributed according to a GMM, the LMM described in (4) will also follow a GMM distribution [11]. This result further motivates the attempt to describe the entire data by a GMM structured statistical model.

A drawback of using a GMM is the difficulty in fitting the data well when it is high dimensional, such as in HSI. This issue can be resolved to some extent by means of dimensionality reduction [11] [12], taking into account the redundancy of information which usually exists in hyperspectral data. The principle component analysis (PCA) algorithm can be very useful for reducing dimensionality while preserving the majority of information present in the original data. For detection problems however, such methods of dimensionality reduction impose a significant problem, since the information on rarely occurring spectral signatures (such as a requested target in many detection problems) will not necessarily be preserved after truncation [9]. Target detection scenarios indeed require alternative methods for overcoming the problem of dimensionality when fitting a semi-parametric distribution.

3. Methods for Evaluation:

Having estimated the distributions and missing parameters and substituting into (1), we obtain a complete test statistic for any given pixel. To quantify the capabilities of the presented detector, we use an exhaustive empirical characterization technique, presented in [16], which utilizes any given dataset and is based on receiver operating characteristic (ROC) curves.

We run the algorithm on the given dataset twice, setting a different hypothesis to hold true at each run. First we generate target-absent pixel test scores by calculating the test statistic for each pixel in the original dataset. We then implant a target according to the assumed model and calculate the test statistic again, separately for each pixel. This results in two sets of pixel LRT scores, which we use to construct target-absent and target-present score histograms.

Using the histograms, we can calculate a complementary cumulative distribution function (CCDF) and measure the empiric probability of detection (p_d) and the probability of triggering a false alarm (p_{fa}) as a function of the detection threshold. By sweeping over possible threshold values we obtain the ROC curve, which gives the probability of detection versus the probability of false alarms.

To quantify how well the algorithm preforms at different false alarm rates, we use the Area Under Curve (AUC) metric:

$$A_{th} = \frac{A - 0.5 \cdot th^2}{th - 0.5 \cdot th^2} \quad (7)$$

where the numerator is the effective area under the ROC curve (and above the line $P_d = P_{fa}$) and the denominator is the effective area for a perfect step function – an ideal detector. The value th is the maximum allowed false alarm rate, and limits the region of the ROC which is considered for the calculation of the test metric A_{th} .

Performance evaluation was done using hyperspectral data from the Rochester institute of Technology (RIT) database.

4. Algorithmic Implementation

4.1 Target absent and target present distribution modelling

Using the EM algorithm, we wish to fit a GMM distribution to the entire dataset. Under the assumption that the dataset is large enough, and that a target, if present, occupies a negligible portion of the total number of pixels, this process will yield an approximation for $f(\mathbf{x}|H_0)$.

As mentioned previously, an attempt to apply the fitting process to the entire dataset directly, is prone to errors, and is an example of the aptly dubbed *curse of dimensionality*. This is due to the high dimensional nature of hyperspectral data.

The number of parameters to be estimated will be $O(N^2)$ due to the sizes of the covariance matrices, and the PDF values at each point, which will be required for likelihood computations, will be on very small scales (or very large on a logarithmic scale), and thus more susceptible to floating point and rounding errors.

Restricting the components of the fitted GMM to have diagonal covariance matrices reduces the order of estimated parameters to $O(N)$. However, since high correlations between consecutive spectral bands naturally exist in HSI, this restriction means a significantly larger number of components could possibly be required to achieve a well fitted distribution, as demonstrated in Figure 2.

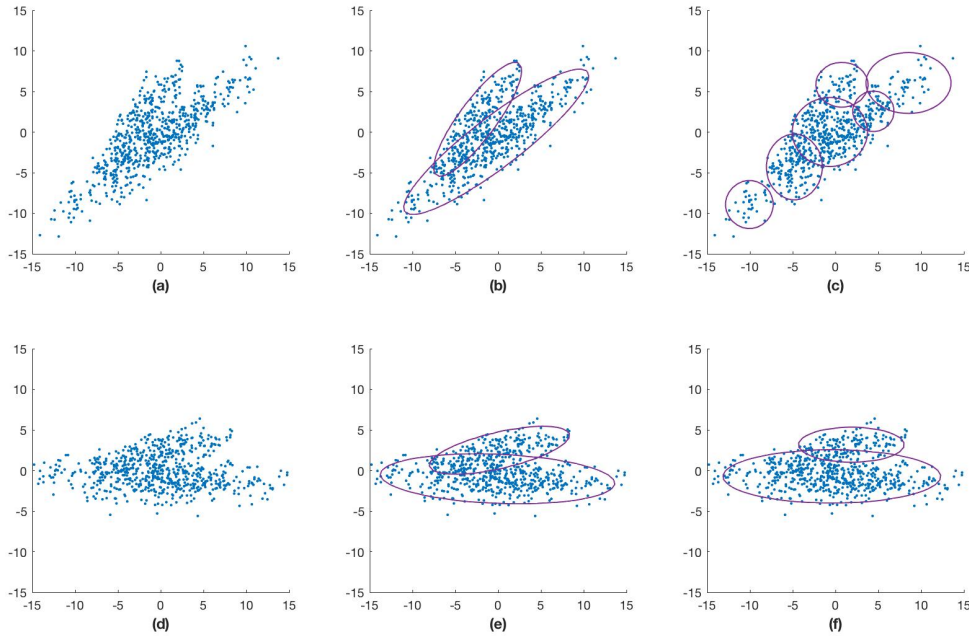


Figure 2: (a) Scatter plot of data points in a 2-D highly correlated feature space. (b) The contours of Gaussian components in a fitted GMM model with 2 components and a negative log-likelihood (NLL) score of 3.6535E3, using full covariance estimation. (c) Contours of Gaussian components in fitted GMM using only diagonal covariance estimation, where 6 components were required to achieve a similar NLL score (3.6846E3). (d) Scatter plot of same data after PCA. (e) Post-PCA data with contours of fitted GMM components using full covariance, with NLL of 3.6535E3. (f) Post-PCA data with contours of fitted GMM components using diagonal covariance, with NLL of 3.6521E3 using only 2 components.

We first whiten the data using PCA, to achieve diagonal covariance. As will be explained in the next part, this step will also allow us to obtain better conditioned covariance matrices for the GMM components.

A transformed pixel is given by $\Psi^T \mathbf{x}$, where Ψ is the PCA transformation matrix, whose columns are the eigenvectors of the data's empiric covariance, arranged by corresponding eigenvalues in decreasing order. Fitting the approximated distribution will be done with respect to the transformed pixels, meaning we are actually approximating $f(\Psi^T \mathbf{x} | H_0)$. For simplicity, we keep the notation $\mathbf{x}$ instead of using $\Psi^T \mathbf{x}$ for the following section. We will use it later when deriving $f(\Psi^T \mathbf{x} | H_1)$, as this derivation is dependent on the exact transformation of the data.

Instead of directly fitting an N-dimensional GMM, we decompose the proposed distribution model into the product of lower dimensional factors. The factorization is based on a probabilistic graphical model (PGM) [17] for the approximated distribution, where each node corresponds to a specific spectral element. We describe the dependencies between the different spectral bands as a Markov network, an undirected graph $G = (V, E)$. In the work presented here, we used a relatively simplistic model, constructing our PGM as independent cliques, meaning the PDF factorizes according to:

$$f_{\mathbf{x}}(\mathbf{x}) = \prod_{c \in cl(G)} \phi_c(\mathbf{x}_c) \quad (8)$$

Where $cl(G)$ are the cliques of G , which are assumed to be mutually disjoint, and $\{\mathbf{x}_c\}_{c \in cl(G)}$ are the corresponding subsets of the elements of $\mathbf{x}$. This representation offers a balance between model simplicity and the ability to capture the dependencies within subsets of the observed features, while performing the actual model fitting in lower dimensional sub-spaces. We do not make a claim of optimality for the proposed decomposition, but will demonstrate its positive capabilities in a detection problem framework.

Since the elements of the post-PCA data are uncorrelated, the covariance of a subset of elements corresponding to a specific clique will also be diagonal and include the corresponding eigenvalues. Since the elements are now arranged according to these eigenvalues in decreasing order, the best conditioned covariance matrix for a subset of elements of a given size will be obtained when the elements are consecutive. An ill conditioned covariance may result in ill conditioned covariance matrices for Gaussian components in the fitted mixture.

We will later compute a maximum likelihood estimate of the fill factor P using the log-likelihood function induced by this PDF estimation. The solution is given by a numeric approximation, so a well condition covariance is a required property. The proposed decomposition will thus be of the form:

$$f_X(\mathbf{x}) = \prod_{i=1}^M f_{X_{S_i}}(\mathbf{x}_{S_i}) \quad (9)$$

Where $\{S_i\}_{i=1}^M$ denote subsets of consecutive elements of $\mathbf{x}$, such that $\mathbf{x}$ is the ordered concatenation of $\{\mathbf{x}_{S_i}\}_{i=1}^M$. The cardinalities of the subsets $\{S_i\}_{i=1}^M$ are selected according to the ratios between the eigenvalues of the complete covariance, given a predetermined tolerance for the condition number of any marginal covariance and a limit on the dimensionality of any fitted GMM. The marginal distributions $f_{X_{S_i}}(\mathbf{x}_{S_i})$ are fitted a GMM model using the EM algorithm, with the model order determined using the Akaike Information Criterion (AIC) [18]. The approximated distribution is given by their product, in accordance with the proposed model.

We now proceed to the matter of deriving the target-present density $f(\mathbf{x}|H_1)$. Examining the assumed replacement model (4), we see that the relationship between a pixel $\mathbf{x}$ under the target-absent hypothesis (H_0) and the target present hypothesis (H_1) is an affine transformation, as we have:

$$H_0: \mathbf{x} = \mathbf{b} \quad (10)$$

$$H_1: \mathbf{x} = P\mathbf{t} + (I - P)\mathbf{b} \quad (11)$$

To derive $f(\mathbf{x}|H_1)$, we could use the following property for affine transformations of random vectors:

Let $\mathbf{Y}$ be a RV with density $f_Y(\mathbf{y})$, and let $\mathbf{Z} = A\mathbf{Y} + \mathbf{b}$, where A is a nonsingular square matrix and $\mathbf{b}$ a deterministic vector. Then:

$$f_Z(\mathbf{z}) = |A|^{-1} f_Y(A^{-1}(\mathbf{z} - \mathbf{b})) \quad (12)$$

Since PCA was applied, we now have:

$$H_0: \mathbf{x} = \Psi^T \mathbf{b} \quad (13)$$

$$H_1: \mathbf{x} = \Psi^T P\mathbf{t} + \Psi^T (I - P)\mathbf{b} \quad (14)$$

In this case, $\mathbf{x}|H_1$ is an affine transformation of $\mathbf{b}$, but not of $\Psi^T \mathbf{b}$, whose density we have estimated under H_0 . Generally, Ψ^T and $I - P$ do not commute. However, since Ψ is orthogonal, we could write:

$$H_1: \mathbf{x} = \Psi^T P\mathbf{t} + \Psi^T (I - P)\mathbf{b} = \Psi^T P\mathbf{t} + \Psi^T (I - P)\Psi \Psi^T \mathbf{b} = \Psi^T P\mathbf{t} + M \Psi^T \mathbf{b}$$

Where $M \triangleq \Psi^T (I - P)\Psi$ is a nonsingular matrix, as for a sub-pixel target, we assume that all entries of P are smaller than 1.

Thus, having previously approximated $f(\Psi^T \mathbf{x}|H_0)$, substituting in (12) we obtain:

$$f(\Psi^T \mathbf{x}|H_1) = |M|^{-1} f(M^{-1}(\Psi^T \mathbf{x} - \Psi^T P \mathbf{t})|H_0) \quad (15)$$

For the additive model, this simply becomes:

$$f(\Psi^T \mathbf{x}|H_1) = f(\Psi^T \mathbf{x} - \Psi^T P \mathbf{t}|H_0) \quad (16)$$

With an expression for $f(\Psi^T \mathbf{x}|H_1)$, we can now use it to compute a maximum-likelihood estimator of P , for an implementation of a GLRT decision rule.

4.2 Fill factor estimation

Having an approximation for the PDF under both hypotheses, and with the aim of computing the GLRT decision rule, we are left with the task of computing an estimate for the fill factor P , which still exists as an unknown parameter in (15) and (16).

We have considered two scenarios. First, a basic case was studied where we assume the factor P is spectrally invariant, and can be considered a scalar as in (2) and (3). In a more general case, we let this factor vary as a function of wavelength. For the latter, we assume this dependency follows some functional model on which we have prior knowledge, but is characterized by a set of unknown deterministic parameters. We have examined several models, all of which include the scalar case as a possible outcome for some specific values of the unknown parameters. Examples of some examined models are presented in Figure 3.

We consider the replacement model as well as the additive model. The approximated density $f(\Psi^T \mathbf{x}|H_1)$ is utilized to compute a maximum likelihood (ML) estimate for either p (when it is assumed scalar), or the unknown constants by which its functional model is parametrized. An expression for the global maximum cannot be explicitly derived from the GMM and is computed numerically, which explains the importance of well-conditioned covariance matrices we attempted to obtain.

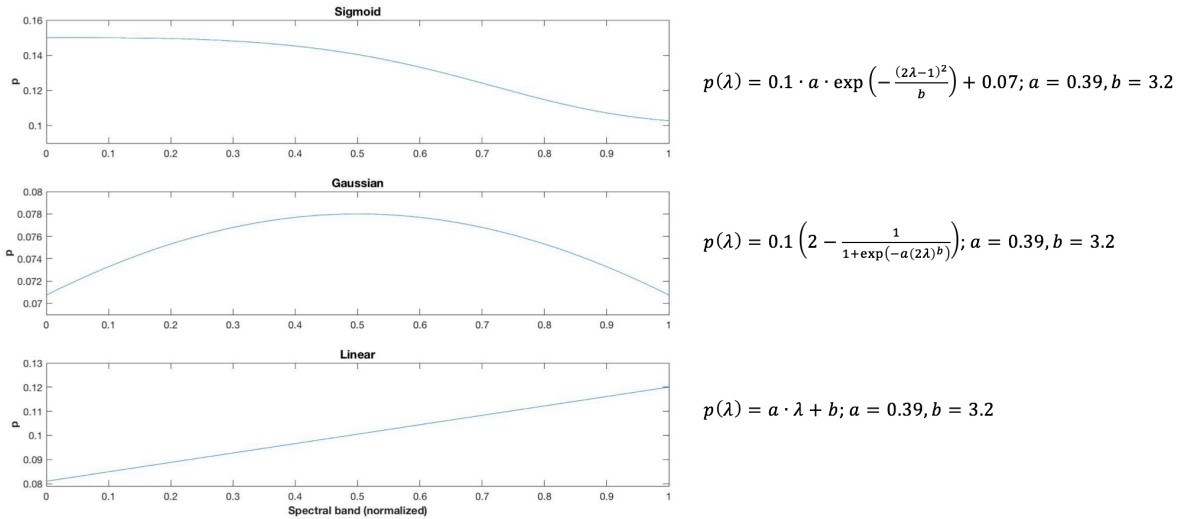


Figure 3: Examples of three different functional models that were considered for the fill factor, each with 2 unknown parameters, denoted a and b . In each figure, the x axis represents the spectrum, normalized to the interval $[0,1]$, and the y axis represents the respective fill factor.

As a reference for the ML estimator, we have also implemented a weighted least squares (WLS) estimator:

$$\hat{p} = \arg \min_p (x - pt)^T W (x - pt)$$

With weighing matrix $W = \Phi^{-1}$, where Φ is the data's empiric covariance. Meaning $\hat{p}$ minimizes the Mahalanobis distance between x and pt , and is given by the explicit expression: $\hat{p} = \frac{x^T \Phi^{-1} t}{t^T \Phi^{-1} t}$.

This scalar estimator is derived optimally when assuming an additive model and attempting to minimize a family of weighted distance functions under a minimum MSE constraint. We therefore expect it to yield good results in a detection scenario with an additive target and constant fill factor. In other scenarios, we would expect the ML estimator to perform better.

5. Test results

We tested our detection algorithm on the RIT blind test database and used receiver operating characteristics to measure and quantify its performance while comparing it to that of the matched filter, which is a commonly used detector in these conditions and is considered a benchmark in many detection problems.

5.1 Spectrally invariant P

Initially we tested our detector under the two basic models given in (2) and (3), where P is a scalar factor, and does not vary with wavelength. Estimation was performed for both additive and replacement models, and using both WLS and ML estimators for P. Our benchmark for comparison is the commonly used matched filter (MF) based detector.

The results displayed below are for a planted target with a power factor of $P = 0.0985$. The histograms for Both ML and WLS estimators are displayed in figure 4. In this case both estimators were unbiased and had similar MSE, but not identically distributed.

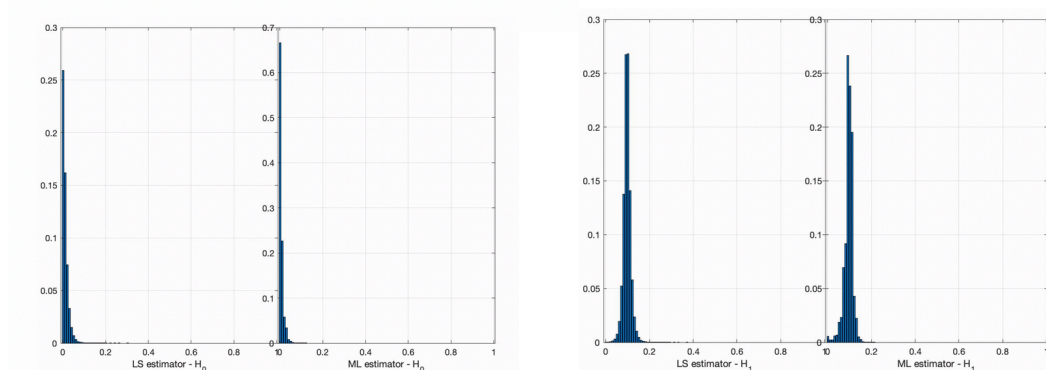
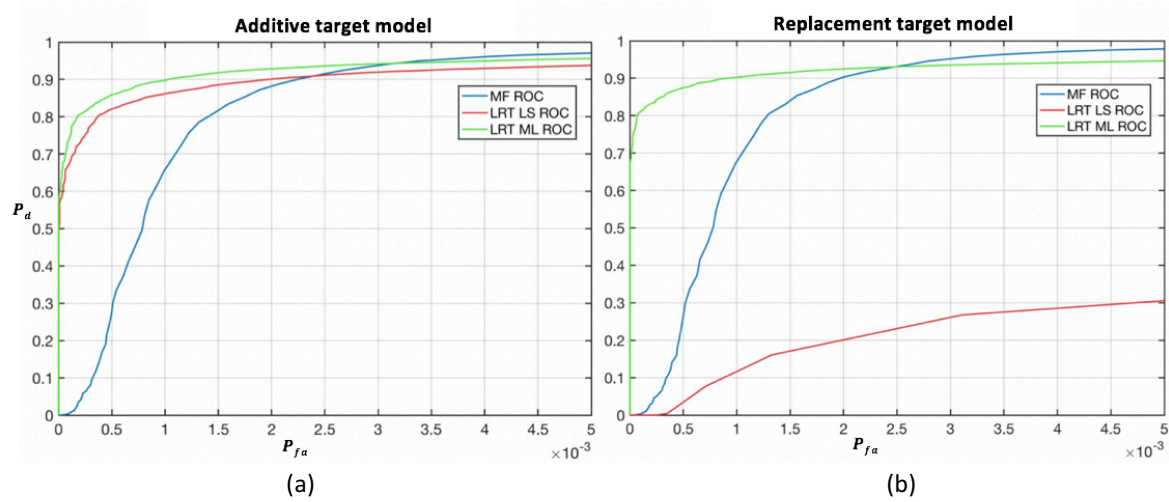


Figure 4: P estimator histograms when actual P used is 0.098, using an additive target implantation. Target absent (left) and target present (right).

As we observe in Figure 5, for low p_{fa} values, the performance of our detector are significantly better than the benchmark, while displaying similar capabilities for higher p_{fa} values. As expected, the detector with a WLS estimator for P performs relatively well while additive target implantation is used, but poorly when the replacement model is applied. The GLRT with ML estimation maintains its high performance for both models, and outperforms the benchmark

for low p_{fa} .



	Additive implanted target		Replacement implanted target	
	$A_{0.005}$	$A_{0.001}$	$A_{0.005}$	$A_{0.001}$
LRT(ML)	0.8675	0.7966	0.8903	0.8404
LRT(WLS)	0.8321	0.7526	0.0915	0.0069
MF	0.7698	0.2769	0.7030	0.2730

(c)

Figure 5: (a) ROC curves for the LRT detector with ML estimator for P (green), WLS estimator (red) and the MF detector (blue) with an additive implanted target using a fill factor of $P = 0.098$. Displayed P_{fa} value range is 0 to 0.005. (b) ROC curves for the three evaluated detectors for a target implanted using the replacement model, with same fill factor, Displayed P_{fa} value range is 0 to 0.005. (c) AUC values for the above ROC curves, for $th = 0.005$ and $th = 0.001$.

5.2 Spectrally varying P

For the next part, we tested the algorithm for the more complex case where P varies as a function of wavelength. We considered several different models which describe how P might vary. In all cases we assumed the functional model is known, and depends on a number of unknown parameters, which we estimate using their ML estimators. As in the previous case, we used a reference WLS estimator. We expect it could perform relatively well under an additive model should the variation in P be subtle enough, but should not perform well under the replacement model.

It is important to note that the MF detector assumes constant P , as does the LRT detector with WLS estimator for P , and therefore we would expect some degradation in detection capabilities in this case for both of these detectors. Figure 7 shows an example of results obtained for one model of P , shown in Figure 6, for both additive and replacement models. In all tested models we observed a similar degradation in the capabilities of the matched filter detector and the WLS estimator (for the LRT detector):

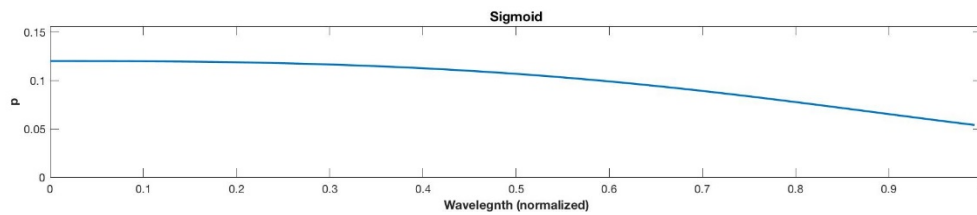
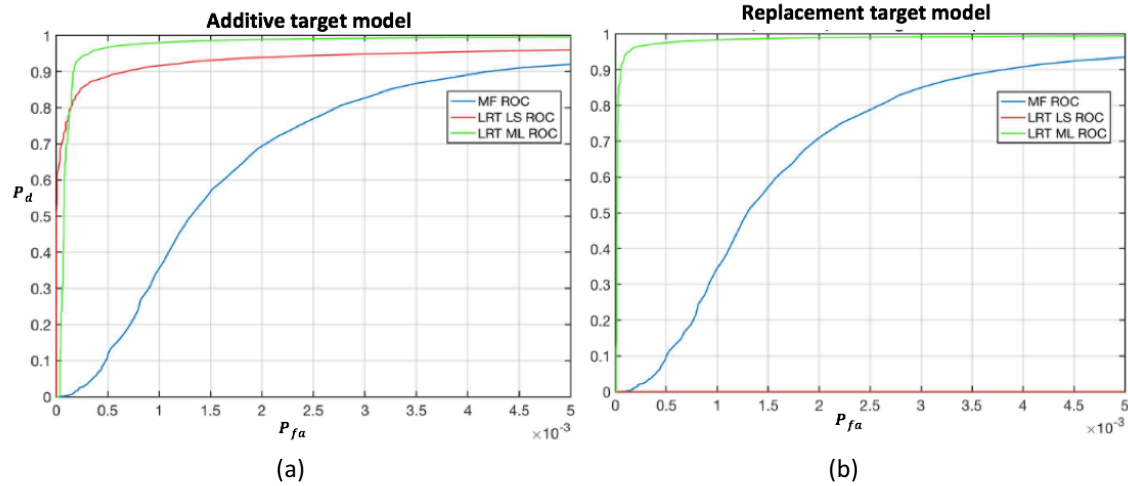


Figure 6: Assumed functional model of P with two unknown parameters a , b . The explicit expression is given by $P(\lambda) = 0.2 \left(1.1 - \frac{1}{1 + \exp(-a(2\lambda)^b)} \right)$, where λ is the wavelength normalized to the interval $[0,1]$.



	Additive implanted target		Replacement implanted target	
	$A_{0.005}$	$A_{0.001}$	$A_{0.005}$	$A_{0.001}$
LRT(ML)	0.8598	0.8220	0.7608	0.7330
LRT(WLS)	0.8524	0.7852	0.0024	0.0013
MF	0.5581	0.1144	0.5329	0.1081

(c)

Figure 7: (a) ROC curves for the GLRT detector, the reference, and the benchmark (MF), with an additive implanted target using the varying P model in figure 6, with parameter values $\mathbf{a} = 0.255$ and $\mathbf{b} = 2.64$. Displayed P_{fa} value range is 0 to 0.005. (b) ROC curves for the three evaluated detectors for a target implanted using the replacement model, with same varying P model and parameter values. (c) AUC values for the above ROC curves, for $th = 0.005$ and $th = 0.001$.

6. Conclusions

The main advantage of the non-parametric model presented in this paper is avoiding the need to make many simplifying assumptions on image background statistics, which most existing algorithms require, and which usually do not actually hold true.

We have consistently observed that for low false-alarm rates ($\sim 10^{-3}$ and lower), which is usually the desired case, the performances of the algorithm presented in this paper, when using the ML estimator for P , are significantly better than the benchmark.

In the more complex scenario, where P varies spectrally, the GLRT detector implemented by this algorithm maintains high performances. The benchmark detector on the other hand shows significant degradation in detection capabilities. In all observed cases, our detector maintains similar performances to those of the benchmark at higher false-alarm rates.

A question that arises from this research is what model of band-varying P should be assumed. It is reasonable to assume that the answer will depend on physical characteristics of the target, instruments and environment, and this will be further explored in future research.

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